

Christian Joudon

SOFTWARE ENGINEER · FULL-STACK · AI / DATA · PRODUCT DESIGN

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SUMMARY

Software engineer who builds full-stack web applications, AI and data pipelines, and fintech systems, usually from design through to deployment. Years of professional creative and design work make for a shared vocabulary with design teams and a faster loop from idea to mockup to working build. Comfortable across React, Next.js, Python, PostgreSQL, REST APIs, and LLM-based language processing.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL, Java, C/C++, R, HTML, CSS, GDScript

Frontend & Design: React, Next.js, Vue.js, Tailwind CSS, Bootstrap, Framer Motion, jQuery, responsive UI/UX, product & visual design, design systems, Adobe Photoshop, Canva

Backend, Data & APIs: Node.js, PostgreSQL, MongoDB, NoSQL, REST APIs, Alembic, database & schema design, ETL pipelines, event-driven architecture, webhooks

AI & Data Engineering: LLMs, NLP, prompt engineering, PyTorch, TensorFlow, Scikit-Learn, spaCy, NLTK, FAISS, semantic search, Apache Spark, PySpark, Databricks, Pandas, NumPy, Pydantic, data visualization (D3.js, Plotly.js, Observable, Matplotlib, Seaborn)

Automation & Integrations: Stripe, QuickBooks, Zapier, Gmail & Drive APIs, Google Apps Script, Clay, Retool, Mapbox, Formspree

Tools & Platforms: Git, GitHub, AWS, Linux, Vercel, Anaconda, Raspberry Pi, FFmpeg, Godot, Meteor.js, Excel, Microsoft Office, Google Workspace

EXPERIENCE

Software Engineer

2025 – Present

HiTech Labs · Kāuaʻi, Hawaiʻi

- Architecting HAVEN, a ledger-first trust-accounting and reservation-management platform, in active agile sprints — with trust-accounting controls, owner-balance protections, and document-linked audit trails on a normalized PostgreSQL data model.
- Built a scalable NLP pipeline that converts unstructured medical encounter notes into clean, structured datasets using LLM-driven text processing and Pydantic validation.
- Integrated FAISS for semantic medication matching and used Apache Spark to analyze comorbidities, medication frequencies, and demographic trends, producing visualizations that informed clinical decisions.
- Developed Chime, a React calendar and booking application with Stripe payments and a fluid, zero-reflow interface built in Tailwind CSS and Framer Motion.
- Integrated payment processing and external APIs (Stripe, AMEX), designed and maintained relational database schemas, and built and deployed responsive production websites end to end.

Accounting Operations & Systems Development

2025 – Present

Hawaii Life · Kīlauea, Hawaiʻi

- Prototyped AI invoice extraction using schema-constrained prompts, validation layers, and human-in-the-loop review, reaching ~87–93% structured-data accuracy.
- Cut manual data-entry time 60–75% by automating file naming, date parsing, and memo formatting through parsing and validation rules.

- Built normalized data schemas for insurance and invoice records and developed secure upload portals with responsive, mobile-first design and automated validation.

Creative Writer

2019 – Present

Freelance · Screenwriting, songwriting & narrative

- Write and develop feature screenplays, short films, 200+ song lyrics (20 released on Spotify), and a children's picture book — building the creative eye and storytelling sensibility that strengthen product and UX work.
- Apply long-form project management and cross-disciplinary communication to bridge creative, marketing, and engineering perspectives.

ADDITIONAL EXPERIENCE

Real Estate Marketing Copywriter · Freelance · 2019–2021 — Property listings, MLS copy, agent bios, and branded digital and print marketing across Zillow, Redfin, and Realtor.com.

Project Manager · Bacchus LLC · 2018–2020 — Managed residential real-estate developments on budget and schedule, coordinating architects, builders, and designers.

Food & Hospitality · Multiple establishments · 2010–2018 — Front-of-house operations, team training and mentorship, and a recipe cost-analysis system that improved profitability.

SELECTED PROJECTS

HAVEN — Full-stack trust-accounting and reservation-operations platform (FastAPI, SQLAlchemy 2.0, PostgreSQL 16, React 19, TypeScript): double-entry ledgers, bank and card reconciliation, platform-payout matching, owner distributions, and month-end close, with financial correctness enforced in the database itself across 200+ org-scoped REST endpoints.

Chime — Embeddable appointment-booking widget (React, TypeScript, Vite, Framer Motion, Stripe): configurable service selection, live availability, and refundable card deposits from a single drop-in component, with graceful payment degradation and a reference Express + PostgreSQL backend.

COVID-19 Records NLP Pipeline — LLM extraction turning unstructured medical encounter notes into clean, validated datasets, with Pydantic validation, FAISS semantic medication matching, and PySpark analysis at scale.

EDUCATION

University of Hawai'i at Mānoa

2020 – 2024

B.S., Computer Science · GPA 3.79 · O'ahu

Relevant coursework: Data Structures, Machine Learning, Web Development, Database Systems.

LANGUAGES

English (Native) · **French** (Proficient) · **German** (Intermediate)

CREATIVE SKILLS

Design: Adobe Photoshop, ad design, pitch deck design, brand and visual design

Photography & Video: professional photography, video editing, photo retouching

Audio: basic sound design, audio editing

Writing & Marketing: creative writing, copywriting, marketing and brand copy, content strategy, email marketing, SEO writing, editing and proofreading, screenwriting, songwriting, storytelling

Presentation & Office: PowerPoint, Keynote, Google Workspace